

# Coulomb + Newton: Towards a Cosmology

Claes Johnson<sup>1</sup>

<sup>1</sup>KTH Royal Institute of Technology, Stockholm, Sweden , [claesjohnson@gmail.com](mailto:claesjohnson@gmail.com)

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## Abstract

RealQM/RealNucleus models nuclei, atoms and molecules as *non-overlapping unit charge densities* — protons carrying the mass  $m_p$  and electrons the mass  $m_e$  — interacting by Coulomb potentials. Being continuum mechanics, the model has the same mathematical form as the models of the macroscopic materials those atoms build, so a hierarchy of continuum models over increasing scales follows from it. Adding **Newtonian gravitation**, which acts on the energy of that matter and matters only in the large, carries the hierarchy to galactic scales: one form of model from nuclei to the cosmic web. The reach is computational and not only formal — both ends are solved by the same primitive, a Poisson solve coupled to a relaxation of a real field on a Cartesian grid, so the gravitational sections are at the same time an application of the small-scale model's computational capacity, with the physics in them deliberately textbook. Cosmology is its natural arena, being the subject in which the nuclear scale meets the astronomical.

The chain is computed throughout. A small perturbation of the electric potential  $\varphi_E$  has its curvature read as charge, quantized by indivisibility into units of  $\pm 1$  in two spatial extents — the compact heavy proton,  $m_p$ , and the diffuse light electron,  $m_e$ . Coulomb builds nuclei, atoms and molecules from them. Newtonian gravitation then acts on the resulting energy density through its departure from the mean,  $\nabla^2 \varphi_g = 4\pi G(\rho - \bar{\rho})$ , a source of both signs although the matter is positive throughout, so the large scale separates into the overdense **cosmic web** and the underdense **voids**. The background source vanishes and the expansion **coasts**,  $a \propto t$ ,  $q = 0$ . The web simulations are run in a periodic box without expansion, so they establish that Newtonian self-gravitating gas makes filaments, sheets and voids from a suitable seed — not that this model produces the observed web. Only at the end is it recorded that the two forces are one two-valued Poisson form read with opposite signs; that parallel is used nowhere in the construction. Neither dark component is accounted for, and the large-scale inhomogeneity is an input.

**Keywords:** RealQM; RealNucleus; two-valued Poisson; Coulomb and Newton; charge quantization; density contrast; gravitation.

## 1 Introduction

RealQM formulates quantum mechanics as multiphase 3D continuum mechanics — unit-charge densities on non-overlapping domains separated by free boundaries, minimising the Coulomb energy [2], developed in the treatise *Real Quantum Mechanics* [1]. Its nuclear extension RealNucleus revives the pre-1932 proton–electron picture (neutron = proton + electron) and reproduces nuclear binding from pure Coulomb interaction, with no strong-force postulate [3]. Because these are continuum-mechanical models — densities on domains, forces from potentials, free boundaries where phases meet — they have the same mathematical form as the models of bulk matter, and a hierarchy over increasing scales follows with no change of mathematical character on the way up. Standard quantum mechanics, formulated in  $3N$ -dimensional configuration space, has no such continuum limit handing directly to the mechanics of materials.

Seen in a longer perspective this is a development of the pre-quantum picture of the atom as a small solar system — one mechanics for the small and the large, in ordinary three-dimensional space — but in new clothes, and the change of clothes is the point. That picture failed because orbiting point charges radiate and spiral in, and because discrete spectra cannot be had from orbits. Here there are no orbits: an electron is a charge density occupying its own domain, in equilibrium rather than in motion, its extent fixed by minimizing energy and not by an orbital radius. What is kept is the ambition, that one form of model should serve at every scale; what is dropped is the feature that made the ambition fail. The present paper carries that form as far as galactic scales, where the governing law is Newton's.

This paper carries that hierarchy to its top. **Newtonian gravitation** acts on the energy of the matter RealQM builds and matters only in the large, so adding it extends the same model to galactic scales — a seamless extension, in that nothing about the mathematics changes: one more Poisson equation, one more force term in the same continuum mechanics. The hierarchy also extends *sideways* at the small scale, and the two directions should be distinguished. Coulomb is the static limit of **Maxwell**, and restoring the time dependence gives **Maxwell + Newton**, which brings **radiation** into the same framework — light as something the matter does (Section 7). This paper stays with Coulomb + Newton and notes where the wider pair would be needed. Where the extension leads is cosmology, and cosmology is its natural arena, being the subject in which the nuclear scale meets the astronomical. Stars shine by nuclear fusion: a process at  $10^{-15}$  m sets the luminosity and chemical output of objects at  $10^9$  m and thereby the evolution of galaxies at  $10^{21}$  m. Light-element abundances are fixed by nuclear reaction rates yet read off the sky as cosmological data. And the dark sector, though *posed* at galactic scales, is *answered* at the smallest — dark matter sought as a new particle in particle colliders and underground detectors — so the instrument aimed at the largest missing mass in the universe is a machine that collides protons. A model that spans from nuclei to galaxies is therefore not an ambition imposed on the subject but what the subject already requires.

**Why gravitation is here, and why it is deliberately textbook.** A reader who reaches Section 9 will find nothing there that Newton did not have: the Poisson equation, a body force, a compressible gas. That is intended, for two reasons worth stating rather than leaving to look like a gap.

The first is that cosmology is not only the large scale. It is the whole chain, and its first question — where the protons and electrons came from — is a *small-scale* question, which is where this framework has something new to say: charge quantized by indivisibility, the neutron as  $p + e$ , nuclear binding from Coulomb geometry, two extents giving exactly two masses. Gravitation's job is to take that matter and gather it. If the gravitational sector were also novel, any success at large scales could be attributed to the new gravity rather than to the matter, and nothing would be learned. Keeping Newton textbook is what makes the test clean: **Cosmology = Coulomb + Newton**, with the novelty in what gravity acts *on*, and gravity itself the control rather than the claim.

The second is that the gravitational sections are a direct application of the *computational* capacity the small-scale model is built on, and are included as a demonstration of it. Nothing in that machinery is particular to Coulomb: it is a Poisson solve coupled to a relaxation of a real, non-negative field on a uniform Cartesian grid, with free boundaries located by the flow instead of imposed. Change the sign convention of the kernel and relax in time rather than downhill in energy and the same primitive is self-gravitating gas dynamics (Section 8). The claim being demonstrated is therefore not that gravitation needs revision but that one computational form, unchanged in its essentials, spans from a nucleus to a cosmic web — close to forty orders of magnitude in length — and does so on hardware as ordinary as a laptop or a browser. A method that could reach only the scale it was designed for would be a far weaker instrument.

**The main idea, and the plan.** *One small perturbation of one potential, followed upward by computation in one form of model, yields quantized charges and masses, then nuclei, atoms and molecules, and finally a web-like cosmos — and every step is computed, not assumed.* The continuity of that chain is the point, more than any one of its links, and the paper follows it in the order the model builds it. Section 2 states the exact model, both tiers. Sections 3–4 start the chain with the perturbation of  $\varphi_E$ ; Sections 5–7 hand the resulting protons and electrons to RealQM and RealNucleus; Section 8 discretises both tiers; Section 9 adds gravitation on the energy density of the matter built; Sections 10–12 ask what more a cosmology would require — more than is delivered here. Only at the end (Section 13) is the parallel between the two forces recorded, as an observation about the finished construction. Security decreases along the chain and is marked as it does: the seed is a posit, the large-scale inhomogeneity an input, the cosmological claims weakest of all. Nothing along the way is static: the equilibria are balances maintained by motion, and the large-scale sector coarsens without ever settling.

## 2 The exact model

The macroscale model is **small-viscosity** (or **vanishing-viscosity**) compressible self-gravitating flow. With  $\rho \geq 0$  the mass/energy density of matter,  $\mathbf{m} = \rho \mathbf{u}$  the momentum,  $e$  the internal energy and  $p$  the pressure,

$$\begin{aligned}\dot{\rho} + \nabla \cdot \mathbf{m} &= 0, \\ \dot{\mathbf{m}} + \nabla \cdot (\mathbf{m} \mathbf{u}) + \nabla p + \rho \nabla \varphi_g &= 0, \\ \dot{e} + \nabla \cdot (e \mathbf{u}) + p \nabla \cdot \mathbf{u} &= 0,\end{aligned}\tag{1}$$

closed by  $\mathbf{u} = \mathbf{m}/\rho$ ,  $p = \gamma_s e$ ,  $e = \rho T$ , and coupled to gravitation through the Poisson equation with the **mean-subtracted** source,

$$\nabla^2 \varphi_g = 4\pi G (\rho - \bar{\rho}), \quad \bar{\rho} = \langle \rho \rangle.\tag{2}$$

Equation (1) is the Euler system with a gravitational body force. Its right-hand sides are zero because no dissipation is put in as a modelling choice; solutions are not smooth — the flow is turbulent and collapse forms shocks — so what is computed is a small-viscosity solution, the viscosity being that of the discretisation, mesh-dependent rather than physical. The meaningful output is therefore what is *stable* as that viscosity is decreased. This is a demanding criterion and we apply it to our own claims: it is what shows, in Section 9, that the arrest of collapse we appear to see is not yet established — the peak density keeps rising as the mesh is refined. Because  $\rho$  stays positive,  $\mathbf{u} = \mathbf{m}/\rho$  is regular and no density floor or clamp is needed anywhere — unlike a genuinely signed density, for which  $\mathbf{u}$  is singular at every zero crossing, precisely at the interfaces where the structure lives.

The  $\bar{\rho}$  in (2) makes the gravitational source take both signs although the matter is positive throughout. It is not an added assumption:  $\varphi_g$  **exists only if its source has zero mean**, so some constant must be subtracted, and  $\bar{\rho} = \langle \rho \rangle$  is the simplest such constant (Section 9).

The small-scale model is the Coulomb energy of RealQM: one non-negative density  $\psi_i$  per electron, each on its own domain  $\Omega_i$ , the domains non-overlapping,  $\Omega_i \cap \Omega_j = \emptyset$ , separated by *free* boundaries. The state minimises

$$E[\{\psi_i\}, \{\Omega_i\}] = \sum_i \int_{\Omega_i} \left( \frac{1}{2} |\nabla \psi_i|^2 - \sum_a \frac{Z_a}{|x - x_a|} \psi_i^2 \right) dx + \frac{1}{2} \sum_{i \neq j} \iint \frac{\psi_i^2(x) \psi_j^2(y)}{|x - y|} dx dy, \tag{3}$$

subject to  $\int_{\Omega_i} \psi_i^2 = 1$ , the domains free to move: the interfaces settle where the energy says. The sum  $i \neq j$  carries **no self-interaction** term — its role is taken by the localisation energy  $\frac{1}{2} \int |\nabla \psi_i|^2$  — and the problem lives in **three** dimensions however many electrons there are, one field each. Both (1)–(2) and (3) are discretised on the same kind of grid in Section 8.

### 3 The seed: charge and mass from a perturbation

In the standard account the origin of matter is an initial hot dense state, with inflation supplying the fluctuation spectrum. Here that role is played by a **small perturbation of the electric potential**  $\varphi_E$ , magnified by the Laplacian  $\rho_E = -\varepsilon_0 \nabla^2 \varphi_E$  into  $\pm$ charge and hence into protons, electrons and all Coulombic matter. That is all the seed is asked to do. It creates the *matter*; it is not asked to lay out the large-scale *distribution* of that matter, and the gravitational sector makes no use of it.

The seed's amplitude at the creation scale is fixed rather than chosen. If  $\varphi_E$  oscillates there with wavelength  $\lambda$  and amplitude  $A$ , each half-wavelength is a domain of one sign carrying charge  $q \sim \varepsilon_0 A \lambda$ ; demanding  $q = \pm e$  exactly — one indivisible quantum per oscillation — gives

$$A \sim \frac{e}{\varepsilon_0 \lambda}. \quad (4)$$

At  $\lambda \sim 1 \text{ fm}$  this is  $\sim 2 \times 10^7 \text{ V}$ , of the same order as the Coulomb potential  $e/4\pi\varepsilon_0\lambda \approx 1.4 \times 10^6 \text{ V}$  of the single charge it creates. The small-scale sector therefore reduces to **one unknown length**  $\lambda$ , the same unknown as the proton size (Section 14).

**And with the charges, the masses.** A cloud's energy is fixed by its extent (localisation cost against Coulomb attraction) and its inertia *is* that energy,  $E/c^2$  (Section 7), so the two extents deliver **two inertial masses** and only two: the compact positive cloud is the heavy one,  $m_p$ , the diffuse negative one the light one,  $m_e$ . Quantization of charge carries quantization of mass with it — exactly two stable species because exactly two extents, so no intermediate mass appears any more than an intermediate charge does. Not settled here is the quantitative map: the observed  $m_p/m_e \approx 1836$  is not reproduced by the naive inverse-size relation, which misses the proton's compactness by about a factor of ten. The *count* of masses is a result; their *values* are open (Section 14).

**Two clarifications.** The seed replaces the Big Bang as the origin of *matter*, not as a beginning of time: there is no initial singularity, and equally no account of why the seed is there or how it evolves —  $\varphi_E$  has no equation of motion here. And the large-scale distribution of matter is not derived from it: a statistically homogeneous seed leaves  $\rho$  uniform up to counting noise ( $\delta \sim N^{-1/2} \sim 10^{-34}$ , thirty orders of magnitude short of what structure requires), so the large-scale inhomogeneity is an **input** to the gravitational sector, not an output of the electric one. Whether one seed could supply both — which would need its long wavelengths to modulate its short-wavelength power, i.e. local-type non-Gaussianity — is left to Section 14.

### 4 The arena

The equations are written in a **3D Euclidean coordinate system** — one of many possible, and nothing depends on which. It is a system of coordinates in which to express the fields, not an absolute space and not a physical medium; no ether is postulated and no frame is privileged. The only operator needed is the Laplacian. Charge is not postulated separately but is the curvature of the potential,  $\rho_{\text{charge}} = -\varepsilon_0 \nabla^2 \varphi_E$ , with two immediate consequences. **Both signs:**  $\rho_{\text{charge}}$  is positive where  $\varphi_E$  is concave, negative where convex. **Net zero:**  $\int \nabla^2 \varphi_E dV$  is a surface term that vanishes, so the universe carries zero net charge automatically — equal numbers of protons and electrons, with no baryogenesis, and once the neutron is read as a proton–electron pair, in the far stronger *local* form of Section 5. Because the Laplacian multiplies each Fourier mode by  $k^2$ , a short-wavelength oscillation becomes a large charge density.

## 5 The small scale: RealQM and RealNucleus

These results are established in [2, 3] and in the treatise [1]; they are *used* here rather than re-derived, and what follows states what the small scale supplies to the chain and on what terms. A reader who grants them may proceed to Section 9. The seed has delivered charge in both signs, quantized at  $\pm 1$ , in two spatial extents: small dense positive and large diffuse negative charge, the **proton** and the **electron**. From here the whole charged world is built by (3), by Coulomb geometry alone.

**Energy, not mass.** In RealQM a particle is a charge cloud whose size is fixed by localisation cost against Coulomb attraction, and its inertia is its energy content  $E/c^2$  — a *measured* identity (the Penning trap, the nuclear mass defect), not a relativistic postulate [4]. No primitive “mass” is invoked: charged matter carries energy, and energy gravitates. A contracted cloud costs more, so small clouds are energy-dense — the small proton is the heavy one. Whether size fixes energy *quantitatively* is another matter, left open; the cosmology needs only that gravity couples to energy.

**Self-repulsion is kinetic energy.** RealQM clouds do not overlap and carry no self-interaction. The role self-repulsion would play, resisting collapse to a point, is carried by the localisation energy  $\frac{1}{2} \int |\nabla\psi|^2$ : compressing a cloud sharpens its gradients and costs kinetic energy, so it settles at finite size. A free electron spreads; a bound one takes the size where localisation balances attraction.

**Charge is quantized:  $\pm 1$  the only possibility.** To exclude an electron’s Coulomb energy with *itself*, the “self” must be well defined: there must be an *indivisible* entity whose interior is left out of the Coulomb sum. A divisible continuum would make “self” ambiguous — is half a cloud a self? do the halves repel? — and the prescription ill-posed. So removing self-repulsion *requires* indivisible units. A unit is identified by its spatial occupation, and for all electrons, and equally all protons, to be the same entity, each domain must carry the same charge. Every electron then carries  $-1$  and every proton  $+1$ , and no intermediate charge can exist. Granted that the no-self-interaction law is one *universal* law rather than an ad-hoc per-blob subtraction, quantization is a consequence, not a postulate — where the Standard Model must *import* it, anomaly cancellation fixing only the  $\pm$  ratio and true quantization needing a Dirac monopole or a compact unifying group, both unobserved. What this does not fix is the *magnitude* of the unit, nor the fractional charges of confined quarks, outside RealQM’s no-substructure scope.

**Nuclei and atoms, one mechanism at two scales.** With a neutron taken as  $p + e$ , a neutral atom  $(A, Z)$  holds  $A$  protons and  $A$  electrons:  $A - Z$  *compact* inside the nucleus and  $Z$  *expanded* in orbitals. The electron occupies two size-phases, its extent adapting to the well: compact in a deep one (nuclear,  $\sim \text{fm}$ ,  $\sim \text{MeV}$ ), expanded in a shallow one (atomic,  $\sim \text{\AA}$ ,  $\sim \text{eV}$ ). Chemistry and nuclear physics are then one Coulomb free-boundary mechanism at two scales: a molecule is nuclei glued by expanded electrons, a nucleus is protons glued by compact ones. RealNucleus reproduces nuclear binding this way — the He-4/deuteron binding ratio comes out 13.1 against an experimental 12.7, from Coulomb geometry alone [3] — and light-nucleus fusion appears as a rearrangement of protons and electrons, not the action of a separate force.

**Neutrality is local.** Because a neutron *is* a bound proton–electron pair, every neutron carries its own compensating charge: matter is neutral *pointwise*, not merely on average. Here that is automatic twice over — globally from the net-zero seed, since  $\int \nabla^2 \varphi_E dV$  is a surface term that vanishes, and locally because a neutron cannot unbalance a count of  $\pm$  pairs.

**What this sidesteps, and what it does not.** It is worth being exact about the comparison, because the corresponding problem in the standard account is a hard one. There, matter originates in a hot plasma of particles and antiparticles, and its survival requires **baryogenesis**: Sakharov’s conditions — baryon-number violation,  $C$  and  $CP$  violation, and departure from thermal equilibrium — must all be met in the early universe. The Standard Model contains all three in principle and fails on two quantitatively. The  $CP$  violation available in the CKM matrix is smaller than required by something like ten orders of magnitude, and for the observed Higgs mass the electroweak transition is a *crossover* rather than first-order, so the departure from equilibrium is not there either. The observed baryon-to-photon ratio  $\eta \approx 6 \times 10^{-10}$  is consequently *not* accounted for by the Standard Model, and supplying it is among the standard motivations for physics beyond it.

In the present picture no such epoch is posited: charge appears in  $\pm$  pairs as the curvature of one potential, so the balance is a property of the arithmetic rather than a residue left by a near-perfect cancellation, and nothing has to be fine-tuned to one part in  $10^9$  to leave the matter we see. But the sidestep should be named for what it is. The standard problem concerns *matter versus antimatter*, whereas the two signs here are the *proton and the electron*, both matter — so the asymmetry question is **dissolved rather than answered**, and that comes at a price. Antimatter exists: positrons are produced in the laboratory and observed astrophysically. A positive charge cloud of *electron* extent is exactly what a positron would be in this framework, so nothing forbids one, and the model owes an account of why the universe is not populated with them — the counterpart, in this language, of the question it has avoided. We do not have one, and it belongs with the open problems of Section 14.

**The comparison is also computational.** The contrast in machinery is as sharp as the contrast in content — lattice QCD and  $\mathcal{O}(N^7)$  chemistry packages against a Poisson solve and a relaxation on a grid — and it is set out with the rest of the comparison in Section 12, together with the honesty it requires: those codes are complicated because their formulations are, and they buy quantitative results this construction does not reach.

## 6 Complexity, and the two domains

The world is built not on a charge imbalance (charge is exactly balanced) but on a *size* imbalance: the proton has a fixed small size while the electron’s is variable, adapting to its well. The fixed proton supplies anchors; the variable electron *spans scales* — spreading to bind atoms, compacting to bind nuclei — producing the nested hierarchy that *is* complexity. With a single fixed size only one scale exists and nothing complex is built, so the nuclear/atomic separation is a **precondition for complexity**: an anthropic reason why the scales are separated, though not a derivation of how far. Since a neutron is structurally  $p + e$  it is heavier than a proton by construction, so proton and hydrogen stability is automatic — the usual fine-tuning bound “ $\alpha$  must not be too large or the proton decays” does not arise.

**Why the proton must be compact.** A caged electron, held by several protons, sits at zero localisation and takes the proton size  $R_p$ ; a free one takes the atomic size  $a_0$ , whence  $a_0/R_p \approx 1/\alpha^2$ : the two domains barely overlap because the two branches are separated by  $\alpha^{-2} \approx 2 \times 10^4$ . The picture also states its own validity limit. Since inertia is energy, a compact electron bound by Coulomb at the nuclear scale has binding  $\sim e^2/R_p$ , and requiring it to stay below the electron’s own energy  $E_e$  gives  $R_p \gtrsim e^2/E_e$ : a floor under the proton size, a plain comparison of two measured energies needing no appeal to  $\alpha$ . Nature sits essentially *at* the floor — at the deuteron radius ( $\sim 2.1$  fm) the Coulomb binding  $e^2/R_p \approx 0.7$  MeV already exceeds  $E_e \approx 0.5$  MeV — so the nuclear scale saturates the theory’s own validity limit, which is why  $\beta$ -decay is where the static-cloud description is most stretched.



## 7 Inertia is energy; the local speed limit

Throughout, inertia is energy content:  $\mathbf{p} = (E/c^2)\mathbf{v}$ , the measured relation of the Penning trap [4], with  $\mathbf{F} = \dot{\mathbf{p}}$  governing the response. In the small the force is electromagnetic and couples to *charge*; in the large it is gravity and couples to *mass*  $= E/c^2$ . One mechanics, one inertia, two forces reading two properties of the same matter.

The constant  $c$  enters *only* the electromagnetic sector, through the field momentum of a moving charge. Hence a **local speed limit**: feeding energy to a charge only increases its inertia  $E/c^2$  without bound as  $v \rightarrow c$ . Newtonian gravitation is by itself  $c$ -free and appears to impose no such limit on the large-scale motion of neutral bodies, which would allow two galaxies to separate at relative speed  $> c$  as a global relative velocity rather than a local signal. **That reading is withdrawn**, and the reason is observational rather than formal: taken with the Hubble law it yields the classical-Doppler distance relation  $d_L = cz/H_0$ , which is excluded by the supernova data by nearly a magnitude at  $z \sim 1$  (Section 12). The recession must therefore be treated relativistically, with the speed limit applying to the bulk motion of matter and not to the electromagnetic sector alone. What survives here is the narrower and still unusual claim: that the limit is a statement about *inertia growing as energy is fed to a charge*,  $E/c^2$ , rather than a postulate about spacetime.

**Coulomb is the static limit; Maxwell adds radiation.** What is used throughout this paper is *electrostatics* — Coulomb’s law, the  $t$ -independent first member of the electromagnetic pair. That is a restriction of convenience, not of principle, and the rung above is worth naming because it is the one that adds the missing physical process. Replacing Coulomb by the full **Maxwell** equations extends the same continuum framework to time-dependent charge and current, and with them to **radiation**: the model becomes **Maxwell + Newton**, and light ceases to be an ingredient supplied from outside and becomes something the matter *does*. Nothing in the mathematics changes character here either — fields on the same space, sources from the same charge densities — so this is a continuation of the hierarchy rather than a departure from it.

The gain is not decorative. Two of the three epochs of Section 11 are *defined* by radiation: nuclei form and the Coulomb binding energy leaves as light; atoms form and the matter turns transparent, releasing the radiation that had been trapped. In the present electrostatic treatment those steps are described by their energy bookkeeping alone — so much binding energy released — with no account of the emission itself. Maxwell supplies the missing half, and it is also what any eventual treatment of the microwave background would have to be built on. The time-dependent form of RealQM required for this, in which a real charge density carries a *complex* current and a radiating dipole, is developed in [5]; nothing in the present paper depends on it, and we make no claim here about radiation beyond noting where it enters.

## 8 The computational model

Both tiers are solved by the same numerical primitive: a Poisson solve coupled to a relaxation on a uniform Cartesian grid,  $\nabla^2$  the standard 7-point stencil, the matter field then advanced — downhill in energy at the small scale, in time at the large one. No basis sets, no wavefunctions in  $3N$  dimensions, no fitted parameters.

|               | Small scale (Coulomb)                                                                   | Large scale (Newton)                             |
|---------------|-----------------------------------------------------------------------------------------|--------------------------------------------------|
| unknown       | densities $\psi_i \geq 0$ on domains $\Omega_i$                                         | contrast $\rho$ , momentum $m$ ,                 |
| field solve   | $-\nabla^2 \varphi = \rho_{\text{charge}}/\varepsilon_0$                                | $\nabla^2 \varphi_g = 4\pi G(\rho - \bar{\rho})$ |
| solver        | multigrid V-cycles, Jacobi smoother                                                     | red-black SOR, $\omega = 2/(1 +$                 |
| relaxation    | steepest descent on the Coulomb energy                                                  | explicit compressible flow i                     |
| free surfaces | interfaces $\partial\Omega_i$ move with the energy                                      | the $\rho = 0$ contact wall                      |
| typical grid  | one global lattice, $200^3$ for a large molecule; $\sim 10^3$ cells per electron domain | $200^3 = 8 \times 10^6$ cells                    |

**Why one code covers both, and exactly how far the sameness goes.** The two columns are not similar by arrangement. Read the *field solve* row: the two equations are one equation, differing in the sign of the source and in a constant. Coulomb and Newton pose the *same computational problem* — invert a discrete Laplacian on a grid against a two-valued source — so the expensive and nontrivial half of the work is literally one routine, and it neither knows nor cares whether the grid spacing is a femtometre or a megaparsec. That is why there is no separate large-scale method to develop, and why the code stays small while the physics changes scale by forty orders of magnitude. It is also, stated computationally, the observation recorded formally in Section 13; nothing in the construction takes it as a premise, but the solver has been using it throughout.

The sameness stops in a definite place, and it is worth marking. What differs between the columns is the *relaxation*: steepest descent to a static minimiser at the small scale, explicit time-stepping of a compressible flow at the large one. That difference is one of question — equilibrium versus evolution — not of scale or of force, and a static gravitational configuration or a time-dependent Coulomb problem would swap the two without touching the field solve. The choice of iteration (multigrid against red-black SOR) is likewise implementation and history: either solves either, and the pairing above is what each tier was written with, not what it requires.

At the small scale the cost grows *linearly* in electron number where a wavefunction method grows exponentially, and the Neumann-free interface — located by the flow rather than imposed — lets one code describe an atom, a molecule and, with the electron compacted, a nucleus. These solvers run in the browser (WebGPU).

At the large scale, since the model is small-viscosity flow the scheme supplies no physical viscosity: the computation *is* the regularisation, and a result counts only if it persists as that regularisation is weakened. Two numerical points carry physics. First, the velocity is taken from  $|\rho|$ ,  $u = m/(|\rho| + \rho_{\min})$ , so **inertia is positive in both phases** while the sign enters only the gravitational coupling — the direct expression of the contrast reading, underdense regions being ordinary matter of which there is less than average. Taking the velocity from signed  $\rho$  instead produces the classic runaway: the negative phase cannot self-support and is consumed, the void fraction collapsing from 50% to a few percent. Second, transport must be conservative and upwind; with centred differencing the web survives but is over-diffusive and the filaments never sharpen.

**What the computations establish.** At the small scale they are quantitative: binding energies, ionisation energies and nuclear binding follow from the same free-boundary minimisation without adjustable parameters. At the large scale they are *morphological only* — dimensionless units, a periodic box, no expansion, no redshift, no calibration to any survey — and nothing in them tests the cosmological claims of Sections 10–12. As *physics* that is a weak claim, and it is not strengthened here. As a demonstration of the *method* it is a strong one, and the two should not be confused: an  $8 \times 10^6$ -cell self-gravitating flow and a free-boundary atom are the same solver on the same grid at scales some forty orders of magnitude apart, with no basis sets, no fitted parameters and no change of formalism between them.



## 9 The large scale: Newtonian gravitation

Nothing in this section is new, by design (see the Introduction): it is Newtonian gravitation as it has always been, acting on the matter the preceding sections built. What is being carried over from the small scale is not physics but *method* — the same grid, the same Poisson solve, the same relaxation of a real field. The mass/energy density of that matter — protons, electrons, their binding and radiation — is  $\rho$ , positive throughout. It is the source of the second force, but enters its Poisson equation only through its departure from the mean, (2), so the *source* takes both signs — positive where matter is denser than average, negative where sparser — while the matter itself is positive everywhere. There is no negative mass and no second species: the two signs are *excess* and *deficit* of one substance.

**The subtraction is standard, and it is a choice.** On a periodic box  $\nabla^2 \varphi_g = 4\pi G \sigma$  is solvable only if  $\langle \sigma \rangle = 0$ , so a constant must be subtracted there as a matter of arithmetic. On unbounded  $\mathbb{R}^3$  the situation is weaker and should not be overstated: a uniform source does admit solutions —  $\varphi = \frac{2\pi G \rho_0}{3} |x|^2$  is one — but no solution is translation-invariant, the isolated-system boundary condition at infinity cannot be met, and the force on a point depends on the order in which the infinite distribution is summed. That is the Neumann–Seeliger difficulty, and it is a difficulty of Newtonian cosmology as such, not of this model. Standard practice removes it by subtracting the mean and calls it the Jeans swindle; taking  $\bar{\rho} = M_{\text{tot}}/V_{\text{tot}}$  makes the total gravitational charge vanish, matching the zero net charge of the Coulomb sector, and dilutes as  $a^{-3}$  so the cancellation holds at every epoch.

None of that is new, and it should be labelled accurately: (2) is a **modelling choice** — peculiar-potential dynamics on a periodic domain — and *not* a theorem forcing the mean to be non-gravitating. The non-standard step is to take that choice as physics: the mean carries no gravitational charge in fact, no relativistic background is imported to restore it, and the consequences are followed to the end (Section 10). That is where the model parts company with general relativity rather than with Newton, since a homogeneous background there does curve spacetime. What it is *not* is modified gravity: the force law is untouched, and for a galaxy, where the cosmic mean lies thirty orders of magnitude below the local density, subtracting  $\bar{\rho}$  changes nothing. The accurate description is *standard Newtonian dynamics with a non-standard cosmological boundary condition*.

**The force-rule.** Under **like sources attract, unlike repel**, all four combinations are realised and each is observed. Two overdense regions attract: clustering. Two underdense regions attract: *voids merge*, as they are seen to. An overdense and an underdense region repel: matter drains from voids into filaments. The large scale therefore separates into overdense and underdense domains — the luminous **cosmic web** of filaments and clusters, and the **voids**, which are not inert leftovers but push their surroundings outward. The asymmetry with the charge sector is worth marking: a positive charge can be isolated, whereas the two signs here are excess and deficit of one substance, meaningless without the mean to measure against. That is why charge comes in indivisible  $\pm 1$  units carrying an identity while gravitational “charge” is continuous and reference-dependent; the quantization argument belongs to the Coulomb sector and must not be carried over.

**What sets the scale, and what arrests collapse.** The web scale is the **Jeans length**  $\lambda_J = 2\pi c_s / \sqrt{G\rho}$  with  $c_s = \sqrt{\gamma_s T}$ : a box below one Jeans length is pressure-supported and inert. What *arrests* collapse is the adiabatic index  $\gamma_{\text{ad}} = 1 + \gamma_s$  through  $M_J \propto c_s^3 / \sqrt{\rho}$ : under adiabatic compression  $c_s^2 \propto \rho^{\gamma_{\text{ad}} - 1}$ , so  $M_J \propto \rho^{3(\gamma_{\text{ad}} - 1)/2 - 1/2}$ , rising with density when  $\gamma_{\text{ad}} > 4/3$  and falling below it. Above  $4/3$  collapse shuts itself off at finite scale; below, each collapsing region keeps fragmenting. (An earlier version attributed the arrest to an artificial shock viscosity. Numerical

experiment does not support that: the shock-heating term destroys the web in a thermal runaway and worsens rather than converges under timestep refinement.)

**Web formation is standard, and illustrated rather than claimed.** Equations (1)–(2) are ordinary Newtonian self-gravitating gas dynamics with a mean-subtracted source; nothing in them is new. That such systems develop a filamentary web from a Gaussian seed is classical: by the Zel’dovich approximation [8] collapse proceeds anisotropically, since the tidal-tensor eigenvalues generically differ, so a region contracts first to a sheet, then a filament, then a node — the morphology made quantitative by Hessian-eigenvalue classification [9]. Figure 1 shows the model’s own equations doing this, in a coasting background and with  $\gamma_{\text{ad}} = 1.4 > 4/3$ , so that the arrest criterion above is satisfied.

Three things in that run are worth stating, since each was checked rather than assumed. *The arrest is **not** demonstrated, and we say so.* At any one resolution peak/mean does stay bounded near  $10^3$  at  $\gamma_{\text{ad}} = 1.4$ , against  $1.3 \times 10^4$  and climbing for the same seed at  $\gamma_{\text{ad}} = 1.2 < 4/3$  — which looks like the Jeans-mass criterion doing its work. But a resolution ladder on an identical seed at fixed time ( $N = 64, 96, 128, 160$ , the seed Fourier-restricted to each grid so the physical field is the same) gives peak/mean 650, 912, 1232, 1616 — a factor 2.49 for a factor 2.50 in resolution, i.e.  $\rho_{\text{peak}}/\bar{\rho} \propto N^{1.00}$ . The bound is therefore the *mesh*, not the physics. The exponent also says what is happening:  $\rho_{\text{peak}} \propto 1/h$  is the signature of a *sheet* one cell thick — a Zel’dovich pancake whose caustic the grid cannot resolve — and not of a collapsed core, which would steepen far more. Consistently, the Jeans length at the density peak stays at 2–3 cells at every resolution, short of the  $\sim 4$  needed to resolve a collapse, and refining does not improve it because the peak density rises to cancel the gain ( $\lambda_J \propto \rho^{-0.3}$ ).

The volume statistics behave better and are quoted on that basis: over the same ladder the deep-void fraction runs 41.7, 46.5, 49.3, 51.3 per cent, with increments decreasing roughly geometrically towards  $\sim 55$  per cent, and the concentration measure likewise. Morphology is thus good to a few per cent while peak densities are not converged at all. The analytic argument for arrest is untouched by this —  $M_J \propto \rho^{0.1}$  does rise for  $\gamma_{\text{ad}} > 4/3$  — but the exponent is so small that demonstrating it numerically needs resolution we do not have. What the runs do show is the *difference* between the two indices, which is robust across the ladder. *The web is nevertheless a transient:* it forms, sharpens, and then *coarsens*, small nodes merging into large ones until little web is left, so the morphology has a window rather than a limit. And *a non-expanding box is worse, but not for the expected reason.* It builds a web and then destroys it thermally: with no expansion to drain the heat of compression, pressure unwinds the structure and the deep-void fraction falls from 87% back to  $\sim 0\%$  with peak/mean returning to  $\sim 10$ . Expansion is thus not needed to keep the box from collapsing — a zero-mean source exerts no net force on a uniform state, so a static box is self-consistent here in a way it is not in ordinary Newtonian cosmology — but it *is* needed to keep a web from blowing itself apart. The two roles should not be confused.

A caution about reading such runs, and it cuts both ways. Volume statistics and visual impression disagree, and neither is sufficient alone. In the run of Figure 1 the deep-void fraction and the mass concentration both climb monotonically — to 73% and 80% — while the picture is becoming *less* web-like, because concentration keeps rising as the web coarsens into isolated clumps. Conversely a thin bright filamentary image can occupy very little volume and be classified almost entirely void. A further trap is instrumental: a Hessian-eigenvalue classifier thresholded on the *r.m.s.* curvature reports the field as structureless exactly when it is most clumped, since a handful of extreme peaks then set the threshold and genuine ridges fall below it (a median-based threshold does not do this). Web-like appearance, web-like statistics and web-like classification are three different claims.

**What the seed of Figure 1 is, and why it is not innocent.** The seed there is a Gaussian random field with a *single correlation length*, placed above the Jeans scale. That choice is doing

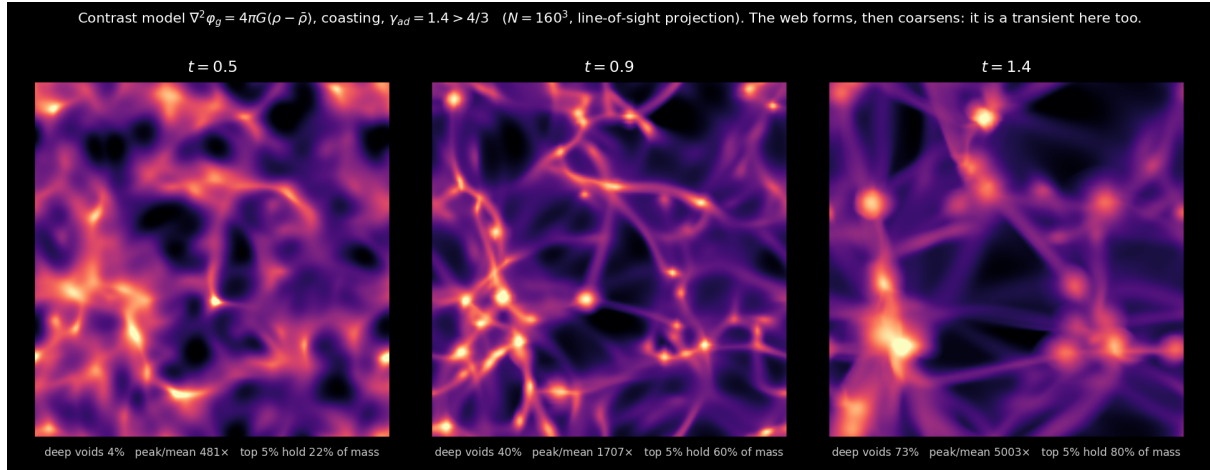


Figure 1: Gravitational instability of the contrast source (2) from a Gaussian seed, coasting background,  $\gamma_{ad} = 1.4 > 4/3$  ( $160^3$ , line-of-sight projection, logarithmic scale; bright = overdense). Filaments meeting at nodes with deep voids between, then coarsening. Total mass is conserved to  $10^{-5}$ . The volume statistics drift by a few percent per refinement and the peak density not at all convergently (see text), so the morphology is indicative and the peak values are upper bounds set by the mesh. The morphology is the classical Zel’dovich sequence [8] and is *not* particular to this model; it illustrates the dynamics and is not offered as evidence for the cosmology.

more work than a figure caption usually admits, and the alternative was tried. Seeding instead with the scale-free  $P_\rho \propto k$  of Section 10 produces *no structure at all*: at equal total variance the run reaches peak/mean 1.8 against 912, with no deep voids, because  $P \propto k$  carries power per logarithmic interval  $\propto k^4$ , so nearly all the variance sits *below*  $\lambda_J$  where pressure forbids growth. Renormalising instead to give the same power *at*  $\lambda_J$  does not rescue it: the required total variance is  $\sim 26$  times larger, which drives the density negative over 46% of the volume, and the seed is then not a perturbation but a saturated field.

That leaves the question the two runs were circling, which is not whether the seed collapses but whether it carries the observed *shape*. The matter power spectrum rises as  $P(k) \propto k^{n_s}$  with  $n_s \approx 0.96$  on the largest scales, turns over, and falls beyond. Measuring  $P(k)$  directly from these runs (spherically averaged in  $k$ -shells, at  $N = 96, 128, 160$ , all at  $t = 1$ ) gives a consistent and unfavourable answer:

| seed                           | $n$ on $k = 2-8$ , seed $\rightarrow$ evolved | $k$ of the peak of $k^3 P$ | peak/mean |
|--------------------------------|-----------------------------------------------|----------------------------|-----------|
| Gaussian bump (Figure 1)       | $-1.5 \rightarrow -1.0$                       | 16–18, stable in $N$       | 870–2900  |
| scale-free, $P_\rho \propto k$ | $+0.9 \rightarrow -0.9$                       | 24–49, wandering           | 1.7–2.0   |
| observed                       | $\approx +1$                                  | —                          | —         |

Neither seed delivers it. The one that collapses has the large-scale slope wrong by about two in the exponent, with too much power on large scales rather than too little, and its web scale is simply its own correlation length —  $k \approx 17$ , fixed by the seed and unchanged as the grid is refined, so the characteristic scale of the pattern is put in by hand rather than selected by gravity. The one with the correct large-scale slope does not collapse at all. A control run with the gravitational coupling set to zero shows in addition that the late-time slope is not the seed slope in either case: pressure smoothing decays the small scales and leaves relatively more power at large ones, an effect gravity partly opposes rather than causes. So a seed cannot simply be handed the observed spectrum and be expected to deliver it downstream. The script is `power_spectrum.py` in the numerics directory.

A raw scale-free spectrum is therefore unusable as an initial condition here, in either direction. That is not a peculiarity of this model — it is why standard cosmology processes the primordial spectrum through a **transfer function**, which suppresses the small-scale power before structure formation begins — but this model has no transfer function (Section 10), and the single-scale bump of Figure 1 is in effect standing in for one. The figure should therefore be read as showing that *these equations make a web from a suitable seed*, not that this model produces the observed web from initial conditions it can justify.

Against the observed web the run gets the voids about right and the rest wrong: at the last panel 73% of the volume is deep void (observed  $\sim 60\text{--}80\%$ ), but filaments and walls hold only  $\sim 5\%$  and  $\sim 3\%$  of the volume against an observed  $\sim 10\text{--}20\%$  and  $\sim 20\text{--}30\%$ . The matter sits in structures too thin.

The honest debt runs the other way. Standard simulations reproduce the observed web using *cold dark matter* in an *expanding* background, neither available here. Baryons alone are the harder case: before recombination they are coupled to the radiation and cannot grow, and gas pressure suppresses structure below the Jeans scale — which is precisely the structure-formation argument *for* dark matter, and among the strongest. That these equations admit webs is no answer to it.

Mathematically the mechanism is a **phase separation** driven by the long-range gravitational kernel rather than by local diffusion — a nonlocal Cahn–Hilliard [15, 16] whose kernel is attractive for like sources, so that coarsening is driven rather than frustrated. One consequence matters observationally: the double well is *not* symmetric, since  $\delta \geq -1$  while overdensities grow without bound, so filaments and voids are not statistical mirrors. That symmetry *would* hold for a genuinely signed mass density, and the asymmetry of real voids counts against that reading.

## 10 Voids, the background, and the spectrum

*Void repulsion is real but is not dark energy.* A region below the mean is a negative source in (2), so it pushes matter outward and grows: voids are actively driven, not merely evacuated. But this is a local, peculiar effect, present in any mean-subtracted gravity, and must not be confused with dark energy, whose defining role is *accelerated expansion of the background*. There the model is explicit and negative: with the mean not gravitating the Friedmann source vanishes,  $\ddot{a} = 0$ , and the expansion coasts ( $q = 0$ ) rather than accelerating ( $q_0 \approx -0.55$ ). **The model does not reproduce what dark energy was introduced to explain.** What it offers instead is that no  $\Lambda$  need be introduced and that the coincidence problem does not arise: the cancellation is exact at every epoch rather than mysteriously near-exact now. Acceleration is *excluded*, not merely absent — both sides of the contrast are the same substance and dilute together as  $a^{-3}$ , so the cancellation holds for all time; acceleration requires  $w < -1/3$  while exact cancellation forces  $w = 0$ .

*Negative mass as a substance: set aside.* An earlier form of this model, and work by others [7, 10, 11], takes the negative sector to be a substance rather than a deficit. We do not follow that route: such a mass would be a species never detected, it would *lens* with the opposite sign whereas void lensing is measured and consistent with ordinary underdensity, and a signed density makes  $\mathbf{m}/\rho$  singular at every zero crossing, forcing regularizations that dominated the result in our own experiments. The deficit reading needs none of them.

*Inflation, and the spectrum.* The Laplacian amplifies short wavelengths,  $\rho_k \propto k^2 \varphi_k$ : an inflation of the density *field*, not the exponential expansion of *space*, which addresses the horizon and flatness problems and is not reproduced here. Within the electric sector this has a tidy consequence — one might expect the  $k^2$  weighting to leave nothing at large scales, and it does not, since in power  $P_{\rho_E}(k) = k^4 P_\varphi(k)$ , so a potential with equal fluctuation power per octave ( $P_\varphi \propto k^{-3}$ , requiring no tuning) gives  $P_{\rho_E} \propto k$ .

It is tempting to read that as delivering the primordial *matter* spectrum, and earlier versions

of this paper did. It does not, and the reason is the same one that runs through Section 3:  $\rho_E = -\epsilon_0 \nabla^2 \varphi_E$  is the **charge** density, and charge screens to neutrality precisely at the scales where the matter spectrum lives. Carrying the result across would require the seed’s long wavelengths to modulate its short-wavelength power — the very link this paper does not assume (Section 14). **The large-scale matter spectrum is therefore an input here, not a result**, its shape included; the simulations of Section 9 impose it. What the calculation above establishes is only that the electric sector carries power at every scale rather than collapsing onto the finest, which is what the  $k^2$  weighting might have suggested.

## 11 The cosmic sequence and the helium fraction

The two sectors exchange energy in both directions: the energy of the charged world is mass and sources gravity, while collapse converts gravitational potential energy into the heat that drives the electromagnetic processes. As contracting matter heats, nuclei form ( $\sim \text{MeV}$ ) and the Coulomb binding energy radiates away as the mass defect; on cooling, nuclei capture expanded electrons into atoms ( $\sim \text{eV}$ ), the matter turns neutral, electromagnetism screens out and radiation streams free. Only then does gravitational aggregation build the web. A bound state survives once the bath can no longer ionise it, so the two electron phases switch on at temperatures set by their binding energies, forcing the order plasma  $\rightarrow$  nuclei  $\rightarrow$  atoms.

| epoch  | process                                              | temperature                               |
|--------|------------------------------------------------------|-------------------------------------------|
| plasma | free $\pm$ charges                                   | $> 10^{10}$ K                             |
| nuclei | compact $e$ captured $\rightarrow$ nuclei            | $\sim 10^{10}$ K ( $kT \sim \text{MeV}$ ) |
| atoms  | expanded $e$ captured $\rightarrow$ atoms; radiation | $\sim 3000$ K ( $kT \sim \text{eV}$ )     |

Table 1: Two condensation epochs, separated by  $\sim 10^6$  in temperature.

**The helium fraction: a consistency, not a prediction.** The electron’s two size-phases — compact (inside a neutron, since neutron =  $p + \text{compact } e$ ) and expanded (atomic) — are interconverted by  $p + e \rightarrow n$  and  $n \rightarrow p + e$ . Three quantities enter and all three are *inputs*: the gap  $E_* \approx 1.29$  MeV between the phases, identified with the neutron–proton mass difference (its absolute size is the open nuclear-scale problem, RealNucleus fixing ratios and not the MeV scale); the transition temperature  $T_* \sim E_*/k \sim 10^{10}$  K; and the freeze-out temperature  $T_f \approx 0.8$  MeV, taken from standard big-bang nucleosynthesis — where it is *derived*, by setting the weak reaction rates against the Friedmann expansion rate. Neither ingredient of that derivation is available here: this model has no weak sector of its own and no Friedmann background, so  $T_f$  is not merely an input but an input this framework has no way to compute. In equilibrium  $n/p$  is the Boltzmann factor  $e^{-E_*/kT}$ , locking at freeze-out; a fraction  $\beta$ -decays back (survival factor  $d$ ) and essentially all the rest end in He-4:

$$\left(\frac{n}{p}\right)_{\text{nuc}} = \frac{(n/p)_f d}{1 + (n/p)_f (1 - d)}, \quad Y = \frac{2 (n/p)_{\text{nuc}}}{1 + (n/p)_{\text{nuc}}}. \quad (5)$$

With  $E_* \approx 1.29$  MeV,  $T_f \approx 0.8$  MeV and  $d \approx 0.74$ :  $(n/p)_f = 0.199 \rightarrow 0.140 \rightarrow Y = 0.245$ , the observed primordial helium. Nothing here is derived; RealNucleus’s one genuine computed output, the He-4/deuteron binding ratio, does not enter  $Y$ . Stated at its true strength, the content is only this: *if* the standard neutron–proton thermal history is imported wholesale, the proton–electron reinterpretation of the two phases does not disturb it, and the familiar helium fraction follows. That is a consistency check on the reinterpretation, and no part of it is a cosmological prediction of this framework.



## 12 Towards a cosmology: what would be required

Shared with the standard account: ordinary matter is Coulombic and gravitation attractive between masses; nuclei and atoms form and later assemble into filamentary structure;  $Y \approx 0.25$  is accommodated. The comparison below is deliberately narrow. Gravitation is textbook here by design, so it is not a point of difference, and where the standard account reaches past Newton — relativistic cosmology, the dark sector, the thermal history — it is reaching outside what this paper claims, which cannot be scored as a disagreement. Two axes remain, and they are the two where the accounts genuinely differ: *where matter comes from*, and *what it takes to compute it*.

| Standard account                           |                                                                                                                                                                                                                                                    | Here (RealQM/RealNucleus)                                                                                                                                                                                                                                    |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>1. The origin of matter</i>             |                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                              |
| origin, and the matter/antimatter question | a hot plasma of particles and antiparticles, whose survival requires baryogenesis: Sakharov's conditions, with CKM $CP$ short by $\sim 10$ orders and the electroweak transition a crossover, leaving $\eta \approx 6 \times 10^{-10}$ unexplained | $\pm$ pairs as the curvature of one potential, so no such epoch is posited and the asymmetry question is <i>dissolved rather than answered</i> — the two signs being proton and electron, both matter. Positrons are then owed an account, and none is given |
| charge quantization                        | assigned; constrained by anomaly cancellation, not derived                                                                                                                                                                                         | a consequence of the indivisibility of a unit cloud: $\pm 1$ the only possibility                                                                                                                                                                            |
| the neutron                                | elementary bound state of quarks ( $udd$ )                                                                                                                                                                                                         | composite: proton + compact electron, so proton stability and local neutrality follow                                                                                                                                                                        |
| nuclear binding, and the masses            | residual strong interaction; Yukawa couplings free, $m_p$ from the QCD scale                                                                                                                                                                       | Coulomb geometry with compact electrons, He-4/deuteron 13.1 against 12.7; two extents of one cloud, the <i>ratio</i> geometric and the absolute scale open                                                                                                   |
| <i>2. What it takes to compute it</i>      |                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                              |
| form of the model                          | a wavefunction for all $N$ jointly, in $3N$ -dimensional configuration space, with operators and probability; no continuum limit handing to the mechanics of the bulk matter atoms build                                                           | one non-overlapping real charge density per electron, a function of position alone: continuum mechanics in ordinary 3D, <i>the same form as the macroscopic models</i> , so the hierarchy continues upward without change of character                       |
| machinery                                  | lattice QCD: Monte Carlo over a 4D path integral, $\sim 10^5$ -line codes, millions of core-hours; chemistry packages with $\mathcal{O}(N^7)$ methods and fitted functionals                                                                       | a Poisson solve + relaxation on a grid, linear in electron number, no basis set and no fitted parameter; the atom solver is 350 lines                                                                                                                        |
| what that buys                             | the hadron spectrum at the percent level, chemical accuracy, and quantitative reach far beyond this domain                                                                                                                                         | one instrument across the whole range — nucleus, atom, molecule, and an $8 \times 10^6$ -cell self-gravitating flow — on a laptop, parameter-free, with nothing in it that can absorb a discrepancy                                                          |

Read the last row in both directions. The standard machinery is heavy because its formulations are, and it buys quantitative results this construction does not reach; simplicity is not accuracy. But it is not mere economy either: it buys range (nothing changes between the tiers but the constant in front of the Laplacian), linear rather than exponential cost, and testability, since with no fitted function to absorb a discrepancy a wrong prediction surfaces as a wrong number. It also buys the paper's own argument — a solver that looks like one for macroscopic matter, and works, is evidence that the microscopic world has the same continuum form.

**On the cosmological constant: fixed, not absent.** In the Newtonian limit of general relativity with  $\Lambda$  the field equation reads  $\nabla^2 \varphi = 4\pi G\rho - \Lambda c^2$ , so  $\Lambda$  enters exactly as a constant subtracted from the source — the form of (2), with  $\Lambda c^2 = 4\pi G\bar{\rho}$ ; Einstein introduced it in 1917 with the same Neumann–Seeliger difficulty in view. What differs is its *status*: free in  $\Lambda$ CDM, with an unexplained coincidence with today's matter density, and here fixed by vanishing total gravitational charge, tracking the matter as  $a^{-3}$ .



**The Hubble diagram, and which one it is.** A vanishing background source gives  $\ddot{a} = 0$  and  $a \propto t$ , and it is worth being careful about what follows, because two different distance laws carry that same expansion rate. Flat Friedmann–Lemaître geometry with  $a \propto t$  — the  $R_h = ct$  model — gives

$$d_L(z) = \frac{c}{H_0} (1+z) \ln(1+z), \quad (6)$$

but it obtains it from *metric expansion*, which this paper does not claim. The kinematics asserted here — matter in inertial motion through a fixed Euclidean space, with redshift a Doppler shift — is instead the Milne situation, and taken with special-relativistic Doppler it gives

$$d_L(z) = \frac{c}{H_0} \left( z + \frac{1}{2} z^2 \right), \quad (7)$$

which differs from (6) at the tenths-of-a-magnitude level over  $0 < z < 2$ . The two must not be mixed, and (6) is recorded here only to be set aside: it belongs to a geometry this model does not adopt.

Which law this framework predicts can be settled, and it is worth doing rather than deferring, because the answer forces a choice. Put the observer at the origin at  $t_0$  with matter in free inertial expansion, so that  $v = r/t$  and  $H = 1/t$ . Light received now was emitted at  $t_e$  from  $r_e = c(t_0 - t_e)$ , while its source moves at  $v = r_e/t_e$ . Taking the redshift as *classical* Doppler,  $1+z = 1+v/c$ , gives  $t_0/t_e = 1+z$  and  $v/c = z$  exactly; the flux carries two factors of  $(1+z)$ , one from the energy per photon and one from the arrival rate, so  $d_L = (1+z)r_e$  and

$$d_L(z) = \frac{c}{H_0} z. \quad (8)$$

This is what the kinematics of Section 7 deliver if the speed limit is imposed in the electromagnetic sector alone, so that recession is unbounded and  $v/c = z$  may exceed one. It is also *excluded*: at  $z = 1$  it gives  $1.0 c/H_0$  against  $\approx 1.53$  for  $\Lambda$ CDM and 1.5 for (7), predicting supernovae some 0.9 mag too bright, which is thirty times the systematic floor of modern compilations.

The alternative is to take the recession as relativistic, in which case the Doppler factor is  $\sqrt{(1+\beta)/(1-\beta)}$ , no source reaches  $\beta = 1$ , and the kinematics are those of Milne with (7). That law is close to  $\Lambda$ CDM out to  $z \sim 1$  and is not excluded by inspection.

*The choice this forces.* One observation does *not* decide it, contrary to what is often supposed: the  $(1+z)$  stretching of supernova light curves follows in both cases, since a receding source emits successive pulses from further away and  $\Delta t_{\text{obs}} = (1+z) \Delta t_{\text{em}}$  by the same factor that reddens the light. What decides it is the distance law, and it decides against the unbounded-recession reading. This paper therefore takes the recession to be relativistic and (7) to be its distance law, at the cost of the remark in Section 7 that two galaxies may separate faster than light: they may not, and the speed limit cannot after all be confined to the electromagnetic sector while the matter obeys a Hubble law. The gain is that the framework now has a distance law of its own rather than a borrowed one, and (6), which belongs to metric expansion, plays no part in it.

## What is not claimed

The model is **Coulomb + Newton**, and radiation is not in it: Coulomb is the static limit of Maxwell. The time-dependent form, in which a real charge density carries a complex current and radiates, exists and reproduces atomic spectra [5]; it is simply not turned on this problem. The microwave background therefore sits in a deferred sector. It plays no role in the construction — nothing above uses it or is calibrated against it, unlike  $\Lambda$ CDM, where it is load-bearing — though a relic field is not foreign to the picture, since the sequence of Section 11 passes through a hot dense phase reached by contraction and ends with radiation streaming free. What is not supplied is everything quantitative: the temperature, the isotropy, the acoustic peaks. Its Planckian spectrum is measured; that the peaks record a last-scattering surface at  $z \approx 1100$  and read off the

$\Lambda$ CDM densities is an interpretation made inside a model, and not inherited here. The standard reading takes the spectrum for thermalization in an optically thick medium; the author’s analysis of blackbody radiation [14] takes it for the signature of a resonant lattice. This paper settles neither. The same deferral covers large-scale-structure statistics and the baryon acoustic feature, galaxy formation, and light elements beyond helium.

### The obstacles inside what *is* claimed

A limited ambition excuses what a paper does not touch, and what needs machinery it has not introduced can wait for that machinery. It does not excuse what falls *inside* Coulomb + Newton and comes out wrong. Three items are of that kind, and the first is derived from the model’s own equations rather than imported from outside.

**Structure cannot grow in a coasting background.** The linear growth equation is  $\ddot{\delta} + 2H\dot{\delta} = 4\pi G\bar{\rho}\delta$ , the source being the perturbation’s own self-gravity, so the mean sets the amplitude even though only the contrast gravitates. In standard cosmology  $\bar{\rho}$  and  $H$  are *locked* by Friedmann,  $\bar{\rho} = 3H^2/8\pi G$ , so source and friction scale alike and  $\delta \propto a$ : the factor  $10^5$  that carries  $10^{-5}$  to order unity. Here they are decoupled —  $\bar{\rho} \propto a^{-3} \propto t^{-3}$  against  $H^2 \propto t^{-2}$  — so the source is subdominant at late times, leaving  $\ddot{\delta} + (2/t)\dot{\delta} \approx 0$  with modes  $\delta = \text{const}$  and  $\delta \propto t^{-1}$ : **growth freezes**. Either the mean must gravitate after all, discarding the coasting prediction, or the contrast must be *supplied* at the required amplitude rather than grown — which would make cosmic structure something the seed specifies rather than something gravity builds, and would then have to carry the observed shape of  $P(k)$  as well. This is the sharpest obstacle between the present framework and a cosmology.

**Which branch is taken, and where the obstruction sits.** The second branch is the one this framework points to. The figure  $10^{-5}$ , which sets how much amplification is demanded, is not a measurement of any seed: it is the microwave anisotropy amplitude read as the primordial perturbation at last scattering, an interpretation this paper does not inherit. Here  $\rho = \nabla^2\varphi_E$  is not a ripple on pre-existing matter, it *is* the matter, structured at order unity from the outset, so nothing needs amplifying by  $10^5$  and gravity’s role is morphology rather than amplitude. The price is that the *shape* of the spectrum becomes the seed’s responsibility, and Section 9 reports that no seed yet discharges it; a second price is that galaxy populations do evolve with redshift, which is evidence of some real growth.

It is worth locating the obstruction, because it does not lie where the new content is. Every link in the chain — Newtonian arena, Neumann–Seeliger subtraction, vanishing background source,  $\bar{\rho}$  decoupled from  $H$  — belongs to the gravitational tier. Charge quantization, the neutron as  $p + e$ , nuclear binding and the two masses are statements about matter, indifferent to how the background expands. In general relativity the mean gravitates, Friedmann relocks source to friction and  $\delta \propto a$  returns; what is surrendered is the coasting background and whatever distance law it turns out to imply, not any result of the preceding sections, and restoring growth restores neither the dark-matter amplitude nor an account of the microwave background.

**The arena is fixed Euclidean space.**  $a \propto t$  in a Newtonian setting is not expanding space but matter in free inertial expansion through it. No expansion of space is claimed here: redshift is a **Doppler shift** of sources in relative motion and  $H_0$  a velocity gradient, with the caveat about the distance law noted above. The computations of Section 8 live in a periodic box with no expansion at all, so what they exhibit is Jeans collapse in a static arena — morphology, and not evidence for the coasting background.

**Two further entries, both within the model’s reach.** There is no extra gravitating component co-located with galaxies, so flat rotation curves and cluster mass-to-light ratios stand where Newtonian dynamics without dark matter leaves them. And  $\rho = \nabla^2 \varphi_E$  is a *definition* rather than a law: with no dynamics for  $\varphi_E$  the relation can be read backwards to yield any  $\rho$  one likes, so the seed is an input here — as the primordial spectrum is an input in the standard account — but without the processing that would connect an input spectrum to an observed one, which is what the test of Section 9 showed to be missing. None of this is addressed here. That is why the paper claims an extension to gravitation and not a cosmology.

**A note on the energy, since it returns the same number.** Ways of removing the obstruction have been canvassed — feeding  $\varphi_E$  so the matter is replenished, or conserving it in an eternal universe with no net expansion — and each needs an ingredient outside Coulomb + Newton, so none is pursued here. One piece of arithmetic survives them and is worth recording. Any such account must balance its energy books, and the zero-sum answer of Tryon [13] — matter created together with its own negative potential energy, costing nothing when  $GM/Rc^2 \approx 1$  — fails here in an informative way: that ratio is  $\frac{1}{2}$  at critical density but  $\approx 0.025$  at  $\Omega_b \approx 0.05$ , since this model has no dark matter. The positive side of the ledger is rest energy, carried at  $10^{-15}$ – $10^{-10}$  m where charge fails to cancel; the negative side is gravitational and global, and too weak by a factor of twenty to forty. **The zero-energy condition asks for some twenty times more gravitating mass than the model contains — the same factor asked for by rotation curves and by structure formation, reached independently.**

### 13 Conclusion: two tiers, one two-valued Poisson form

The construction is complete — a perturbation of  $\varphi_E$  quantized into charges, Coulomb building nuclei, atoms and molecules from them, Newtonian gravitation acting on the energy of what is built — and an observation can now be recorded that was used as a premise nowhere above. Each tier was set up on its own terms; that is what makes the observation worth recording.

The two tiers, arrived at independently, turn out to have *one* structure and to differ in a single sign. Each force is the gradient of a potential obeying Poisson’s equation with a two-valued source,

$$\text{electric: } \rho_c = -\varepsilon_0 \nabla^2 \varphi_E, \quad \text{gravitational: } \rho_m = \frac{1}{4\pi G} \nabla^2 \varphi_g, \quad (9)$$

the force being  $-(\text{charge}) \nabla \varphi$  in each case. The one structural difference is the relative sign, and with it the force law:

|                      | like sources   | unlike sources |
|----------------------|----------------|----------------|
| electric (charge)    | <b>repel</b>   | attract        |
| gravitational (mass) | <b>attract</b> | repel          |

There are thus two independent binary signs — the source  $\pm \nabla^2 \varphi$  and the force  $\mathbf{F} = \pm \nabla \varphi$  — whose four combinations are exactly the four “charges” the two forces couple to: the repulsive-for-like sign is electromagnetism, its  $\pm$  source the  $\pm$  charge of the small scale; the attractive-for-like sign is gravitation, its  $\pm$  source the  $\pm$  contrast of the large. One operator and two signs yield  $\{\pm \text{charge}, \pm \text{mass}\}$ .

That single sign difference generates the two tiers. Because like charges repel, matter neutralises into  $\pm$  pairs and electromagnetism *screens*, staying local and small-scale. Because like masses attract, mass *accumulates* into large-scale structure, and because unlike sources repel, overdense and underdense regions separate into web and voids. The scale separation is thus a *consequence* of the force-sign, not a separate assumption. A second asymmetry — the *direction*

in which the Poisson relation is used — fixes the scale of each force. For electromagnetism the potential is primary and charge is obtained by *differentiating*: the Laplacian is local and magnifies short wavelengths ( $\rho_k \propto k^2 \varphi_k$ ), so a small-scale perturbation is inflated into sharp charges. For gravitation the source is primary and the potential is obtained by *integrating*: the inverse Laplacian is global and smooths ( $\varphi_k \propto \rho_k/k^2$ ), building a broad potential. The same relation run forward makes small-scale electromagnetism and run backward large-scale gravitation — and that is why charge is small and gravity large.

One practical consequence of the shared form deserves recording alongside the formal one, since it is what made the chain computable end to end rather than merely storable: sharing an operator, the two tiers also share a solver. The same Poisson-plus-relaxation primitive on the same kind of grid produces a free-boundary atom and an  $8 \times 10^6$ -cell self-gravitating flow, with no change of formalism and no fitted parameters at either end. Whether the framework is right is decided by the physics; that it can be *followed*, scale by scale and without a change of instrument, is decided by this.

This is a *structural parallel*, observed after the fact, and we resist promoting it into a premise. It does not say gravitation is derived from electromagnetism, or that one seed generates both sectors: the gravitational source here is the matter's own energy density and owes nothing to  $\varphi_E$  beyond the matter it made. Nor does the shared form make the two constants one — whether  $\alpha$  and  $\alpha_G$  reduce is the open question below, and the parallel is a reason to ask it, not an answer. What can fairly be said is that two forces spanning forty orders of magnitude in strength, introduced for entirely separate reasons — one to make matter, one to gather it — were found to be the same equation read with two signs.

## 14 Open problems

1. **The proton–electron size difference**, in three parts of unequal status. (i) *The ratio* is read here as geometric: a caged, zero-localisation electron sits at  $R_p$  and a free one at  $a_0$ , so  $a_0/R_p \approx 1/\alpha^2$ , needing no relativity. This the model addresses. (ii) *The absolute scale* of  $R_p$  — the same datum as the mass ratio and the nuclear energy scale: one open number, not derived here, nor in the Standard Model. (iii) *The sign asymmetry*: why + is the compact sign at all. Pure Coulomb with the Laplacian is charge-conjugation symmetric, so the asymmetry cannot arise from these ingredients; a natural candidate is a *non-Gaussian (skewed)* seed, whose super- and sub-level sets are not mirror images — one sign forming sharp compact pits, the other broad swells — a geometric route from the *statistics* of the seed rather than a new force.
2. **Could the two potentials be one field?** Split one potential by scale,  $\varphi = \varphi_{\text{short}} + \varphi_{\text{long}}$ , reading charge as the short-wavelength Laplacian and mass contrast as the long-wavelength one; the objection that neutral hydrogen has vanishing charge but non-vanishing mass does not apply, the two roles never being evaluated at the same scale. It would need a filter scale fixed by something physical, and the abundance of small-scale structure to track  $\nabla^2 \varphi_{\text{long}}$  — which fails identically for a Gaussian seed and requires the local-type non-Gaussianity wanted in item 1(iii), so one assumption might serve both. Whether a skewed seed's small-scale power correlates with its own long-wavelength Laplacian is a well-posed calculation, not attempted here.
3. **The fluctuation spectrum and a law of motion.**  $\rho_E = -\varepsilon_0 \nabla^2 \varphi_E$  is a definition, not a dynamics:  $\varphi_E$  has no equation of motion, and its spectrum is an input at the small scale just as the matter spectrum is an input at the large one (Section 10). These are two separate gaps and the second is the load-bearing one: the shape as well as the amplitude of the large-scale contrast is supplied, so the model does not predict  $P(k)$ . Closing it would need the seed's long wavelengths to modulate its short-wavelength power, as in item 2 above; short of that,

supplying  $\varphi_E$  with a law of motion would close only the first.

4. **The mass ratio  $m_p/m_e = 1836$ , and why it is not the nuclear scale.** The mass ratio implies a nuclear size  $a_0/(m_p/m_e) \approx 29$  fm, whereas the actual scale is  $\sim 2\text{--}3$  fm; equivalently the size ratio  $a_0/R_p \approx 2.5 \times 10^4 \approx 1/\alpha^2$  exceeds 1836 by a factor  $\sim 14$ . The nucleus is that much more compact than the naive size–mass law allows, because the caged electron is flat, held collectively by several protons. Deriving that factor would make  $m_p/m_e$  fix the nuclear scale, hence  $\alpha$ , and collapse two numbers into one; asserting it incidental would be numerology.
5. **Antimatter.** The  $\pm$  signs of the seed are the proton and the electron, both matter, so the matter–antimatter asymmetry of the standard account does not arise (Section 5). The debt this incurs is the mirror question: a positive cloud of electron extent is a positron, nothing in the construction forbids it, and why such clouds are not abundant is not explained here. Pair production, and the fate of the positrons it makes, lie in the time-dependent (Maxwell) sector rather than the electrostatic one treated here.
6. **The neutrino.**  $\beta$ -decay’s continuous spectrum — the 1930 objection that sank the proton–electron neutron [6]. The “electron expands” step needs a home for that energy and spin; an attempt to have a field carry it came out too weak, so the neutrino is neither derived nor eliminated here.
7. **Precision and scope.** Abundances beyond helium, fluctuation-to-structure statistics, and the extreme regimes (strong-field gravity, the highest energies) lie beyond the Coulomb+Newton scope kept here.

**The count of constants.** Only dimensionless numbers are physical. Stripped to essentials the model carries **three**:

|                                                         |                        |                                                        |
|---------------------------------------------------------|------------------------|--------------------------------------------------------|
| $\alpha = 1/137$                                        | Coulomb strength       | sets the size hierarchy, $a_0/R_p \approx \alpha^{-2}$ |
| $m_p/m_e = 1836$                                        | the two charge species | sets the mass hierarchy                                |
| $\alpha_G = Gm_p^2/\hbar c \approx 5.9 \times 10^{-39}$ | gravitational strength | sets where Newton takes over                           |

The first two are small-scale: atoms, molecules, chemistry and nuclear binding ratios follow from them parameter-free, gravity playing no part. The third governs the transition, and does so at a definite scale — a body can be held up electromagnetically only up to about  $\alpha_G^{-3/2} \approx 1.8 M_\odot$ , beyond which gravity wins — so the smallness of  $\alpha_G$  is not an embarrassment but the very thing that creates the two-tier structure.

Against this the Standard Model carries some nineteen free parameters and standard cosmology adds several more, and three against nineteen is a striking reduction — but only *over the domain covered here*, and that qualification does the real work. The right-hand column of what is *not* attempted (quark substructure and QCD, the particle spectrum, precision QED, the weak sector, anything relativistic or at collider energies) is longer and heavier than what is claimed, and a parameter count is meaningless without the domain it covers: on that measure the Standard Model buys its nineteen a great deal more. What can be said is narrower and still worth saying — *within* atoms, molecules and nuclear binding *ratios* this construction is parameter-free where the standard account is not, and it derives two things, charge quantization and local neutrality, that the standard account must supply.

Three caveats close the account honestly.  $\alpha_G$  is not derived, which is the hierarchy problem and no one has derived it;  $m_p/m_e$  and  $\alpha^{-2}$  differ by the unexplained factor  $\sim 14$  noted above; and whether  $\alpha$  and  $G$  themselves reduce to one is the deeper question this deliberately parallel construction does not settle.



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